

DOMS Recovery Protocol

What delayed onset muscle soreness actually is, what genuinely speeds recovery, and what popular remedies the evidence doesn't support.

Evidence-based approach: Cheung et al., 2003; Dupuy et al., 2018; Machado et al., 2016.

What's Actually Happening

DOMS is not caused by lactic acid buildup — that myth persists despite being debunked decades ago. Lactate clears from muscle tissue within about an hour of exercise ending. DOMS is instead the result of microscopic damage to muscle fibers and connective tissue, concentrated in the eccentric (lengthening) phase of a contraction, followed by a local inflammatory response as the body repairs the tissue.

Soreness typically begins 12-24 hours after exercise, peaks between 24 and 72 hours, and resolves within 5-7 days. A 2003 review by Cheung, Hume, and Maxwell in *Sports Medicine* remains one of the most cited summaries of DOMS mechanisms and treatment strategies, and concluded that no single intervention eliminates DOMS entirely — the goal is managing severity and maintaining function, not preventing it outright.

Section 1 — What Actually Helps

1. Light active recovery

Low-intensity movement (walking, easy cycling) increases blood flow to sore tissue and has consistently shown modest reductions in perceived soreness versus complete rest.

2. Massage or foam rolling

A 2018 meta-analysis by Dupuy et al. in *Frontiers in Physiology* found massage and foam rolling produced small-to-moderate reductions in DOMS and perceived fatigue, most effective in the 24-72 hour window.

3. Adequate protein intake

Roughly 0.3g per kg of bodyweight per meal supports the muscle repair process that resolves the underlying tissue damage.

4. Sleep

Growth hormone release and most tissue repair processes are concentrated in deep sleep — cutting sleep short measurably slows recovery capacity.

5. Gradual training progression

The repeated bout effect means muscles adapt to reduce DOMS severity after repeated exposure to a given movement — the single best long-term strategy is progressing load gradually rather than sudden spikes in volume or new exercises.

Section 2 — Mixed or Overstated Remedies

— Static stretching

A 2016 meta-analysis by Machado et al. in the British Journal of Sports Medicine found stretching before or after exercise has little to no meaningful effect on DOMS severity.

— NSAIDs (ibuprofen, etc.)

NSAIDs can modestly reduce soreness perception, but several studies suggest they may blunt the muscle protein synthesis signal involved in adaptation when used regularly post-training. Occasional use for comfort is reasonable; routine use after every session is not well supported.

— Cold water immersion

Effective at reducing short-term soreness perception, but research (including work by Peake and colleagues) suggests frequent use after strength training may blunt long-term hypertrophy and strength adaptations. Better reserved for competition or high-frequency training blocks than routine hypertrophy work.

— Complete rest

Total inactivity does not speed recovery faster than light movement, and in most studies performs worse than active recovery for restoring function.

Section 3 — When to See a Professional

— Soreness beyond 7 days

Ordinary DOMS resolves within a week. Persistent pain beyond that window warrants evaluation for injury rather than assumed soreness.

— Dark urine after intense exercise

A rare but serious condition, rhabdomyolysis, can follow extreme unaccustomed exertion. Dark urine, severe swelling, or inability to move a limb needs immediate medical attention.

— Sharp, localized pain

DOMS is typically a diffuse ache across a muscle group. Sharp, one-sided, or joint-specific pain is more consistent with a strain or injury and should be assessed separately.

A Simple 72-Hour Recovery Checklist

Use this as a general framework after a hard training session, adjusted for the training history. Consistency across sessions matters more than any single means the same session will produce noticeably less soreness the second to it.

- Day 0 (same day): Hydrate, eat a protein-containing meal within a few hours, prioritize a full night's sleep.
- Day 1: Light active recovery (20-30 min easy walk or cycle). Foam roll or self-massage the affected area for 5-10 minutes.
- Day 2: Continue light movement. Reassess soreness — if it's easing, resume normal training at reduced intensity.
- Day 3: Most DOMS from a typical session should be resolving. Return to full training load if pain-free.

Sources

Cheung, K., Hume, P., & Maxwell, L. (2003). Delayed onset muscle soreness: treatment strategies and performance factors. *Sports Medicine*, 33(2), 145-164.

Dupuy, O., et al. (2018). An evidence-based approach for choosing post-exercise recovery techniques. *Frontiers in Physiology*, 9, 403.

Machado, A. F., et al. (2016). Can water temperature and immersion time influence the effect of cold water immersion on muscle soreness? A systematic review. *British Journal of Sports Medicine*.